

Assignment: Detect Intent-to-Speak from EEG

Objective

Design a system to detect **when a person is about to speak** using **EEG signals from only 2–4 electrodes**.

We define:

Intent-to-speak = EEG activity in the -500 ms to 0 ms window relative to speech onset

This is a **binary classification problem**:

- 0: No intent
 - 1: Intent to speak
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What to Submit (6–8 pages max)

1. Problem Setup

- How you define and label positive vs negative samples
 - How you obtain speech onset (ground truth)
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2. Data Collection Plan

- Experimental protocol
 - Number of subjects and trials
 - Any assumptions you make
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3. Hardware & Electrode Choice

- EEG device type
 - Sampling rate
 - Exact electrode placement (2–4 channels) + justification
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4. Preprocessing

- Filtering, artifact handling
 - How you extract the -500 ms → 0 ms window
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5. Modeling

- At least one modeling approach
 - Input representation and why it should work
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6. Training & Loss

- Training setup (splits, strategy)
 - Loss function and evaluation metrics
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7. Expected Performance (Important)

- Realistic accuracy with 2–4 electrodes
 - Within-subject vs cross-subject
 - Detection latency (how early before speech)
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Evaluation

We care about:

- Clarity of thinking
- Practicality
- Correct assumptions about EEG
- Realistic expectations